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Course/Program: BTech C.S.E SY
Subject: Digital Logic and Microprocessor.

Practical - 02.

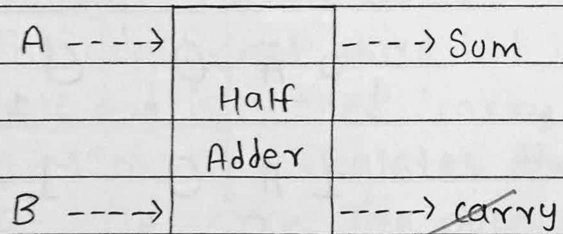
* Aim:

To verify the truth table of half adder and full adder by using XOR and NAND gates respectively and analyse the working of half adder and full adder circuit.

* Theory:

1) Half Adder

→ Half Adder is a combinational circuit that performs simple addition of two binary numbers. If we assume A and B as the two bits whose addition is to be performed, the block diagram and a truth table for half adder with A, B as inputs and Sum, Carry as outputs can be tabulated as follows.



Truth Table			
Input		Output	
A	B	Sum	carry
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

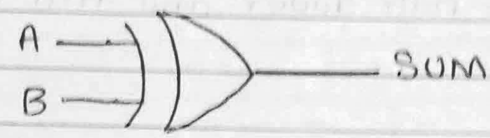
fig: Block diagram and truth table of half adder.

The truth table and K-map simplification and logic diagram for Sum output is shown below:

Truth Table

A	B	SUM
0	0	0
0	1	1
1	0	1
1	1	0

A \ B	0 ($\bar{B}$)	1 (B)
0 $\bar{A}$	0 [0]	1 [1]
1 A	1 [2]	0 [3]



2 input EX-OR

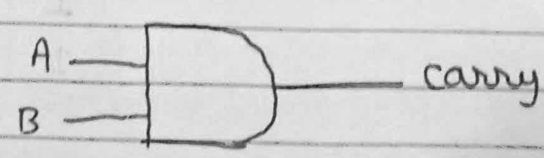
SUM = AB' + A'B

The truth table and K-map simplification and logic diagram for carry is shown below.

Truth Table

A	B	carry
0	0	0
0	1	0
1	0	0
1	1	1

A \ B	0 ($\bar{B}$)	1 (B)
0 $\bar{A}$	0 [0]	0 [1]
1 A	0 [2]	1 [3]



2 input AND

carry = AB

If A and B are binary inputs to the half adder, then the logic function to calculate sum S is EX-OR of A and B and logic function to calculate carry C is AND of A and B. Combining these two, the logical circuit to implement the combinational circuit of half adder is shown below.

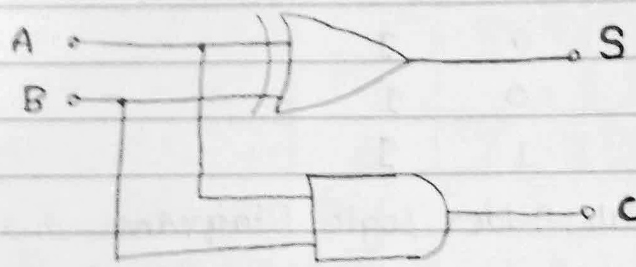


Fig: Half adder logic Diagram.

2). Full Adder.

Full adder is a digital circuit used to calculate the sum of three binary bits. Full adders are complex and difficult to implement when compared to half adders. Two of the three bits are same as before which are A, the augend bit and B, the addend bit. The additional third bit is carry bit from the previous stage and is called 'carry'- in generally represented by CIN. It calculates the sum of three bits along with the carry. The output carry is called carry-out and is represented by carry OUT.

The block diagram of a full adder with A, B and CIN as inputs and S, carry OUT as outputs is shown below.

Truth Table.

Input			output	
A	B	Cin	Sum	carry
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

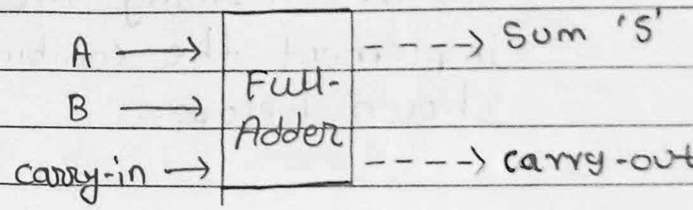


fig: full Adder Logic Diagram.

Based on the truth table, the boolean functions for sum (S) and carry-out (COUT) can be derived using K-map.

A \ BCin	00	01	11	10
0	0	1	0	1
1	1	0	1	0

The K-map simplified equation for sum is

$$S = A'B'Cin + A'B\,Cin' + ABCin$$

A \ BCin	00	01	11	10
0	0	0	1	0
1	0	1	1	1

The K-map simplified equation for COUT is

$$COUT = AB + A\,C_{in} + B\,C_{in}$$

In order to implement a combinational circuit for full adder, it is clear from the equations derived above, that we need four 3-input AND gates and one 4-inputs OR gate for sum and three 2-input AND gates and one 3-input OR gate for carry-out.

* Procedure.

1) Half Adder:

Step: 1) connect the type supply (+5V) to the circuit.
Step: 2) first press "AND" button to add basic state of your output in the given table.
Step: 3) Press the switches to select the required inputs "A" and "B".
Step: 4) Press "AND" button to add your inputs and outputs in the given table.
Step: 5) Repeat steps 3 and 4 for next state of inputs and their corresponding outputs.
Step: 6) press the "RESET" button whenever you want to refresh your simulator.

2) Full Adder.

Step: 1) connect the supply (+5V) to the circuit.
Step: 2) first press "ADD" button to add basic state of your output in the given table.
Step: 3) Press the switches to select the required inputs "A" and "B" and "Cin".
Step: 4) Press "AND" button to add your inputs and outputs in the given table.

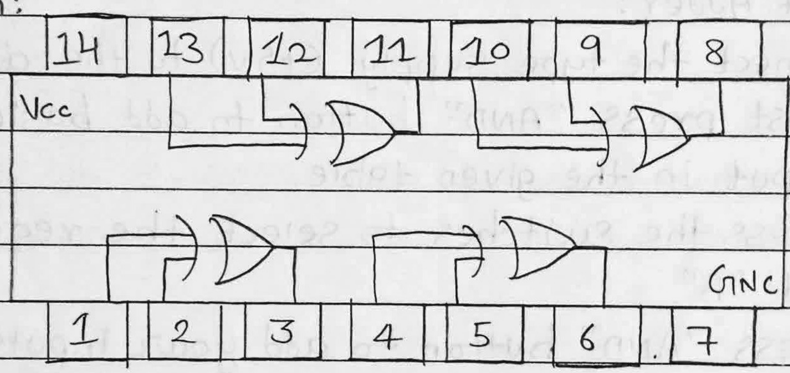
Step: 5) Repeat steps 3 & 4 for next state of inputs and their corresponding outputs.

Step: 6) Press the "PRINT" button after completing your simulation to get your results.

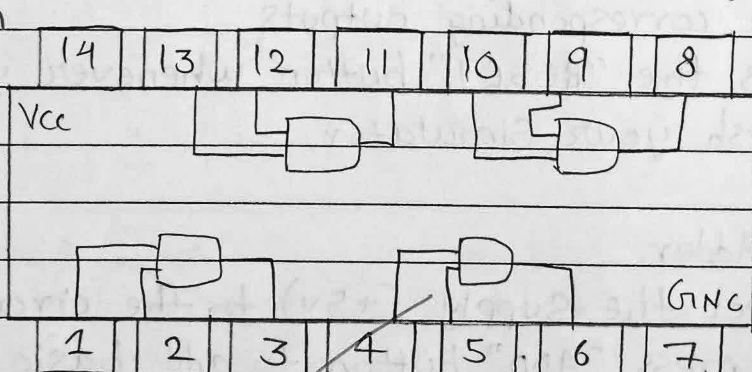
Step: 7) Press the "RESET" button whenever you want to refresh your simulator.

- ICs used in Half Adder Circuit.

1. 7486 XOR gate IC: A quad 2-input logic gate IC
Pin diagram:



2. 7408 AND gate IC: A quad 2-input logic gate IC
Pin diagram



* Conclusion: The experiment validated the truth tables of the half and full adder. The half adder used XOR for the sum and NAND for the carry, while the full adder accurately added two bits with a carry-in. The results matched theoritical expectations, confirming proper circuit functionality.